

remained convinced that if they could draw the US Pacific Fleet into battle, they could destroy the carriers and control the ocean. This was the scenario they envisaged when they invaded Midway Island in June 1942.

Because American cryptographers had cracked Japan's naval codes, the US was forewarned of the Midway operation and so undistracted by a simultaneous Japanese move against the Aleutian Islands. Unable to assemble a naval force comparable to the 200 ships Yamamoto sent into action, Admiral Chester Nimitz, newly appointed commander of the Pacific Fleet, had to depend on his airmen to stop the Japanese invasion. Thanks to heroic efforts by dockyard workers at Pearl Harbor, the *Yorktown* – originally estimated to need 90 days to recover from its battering in the Coral Sea – was repaired in three days, and joined the carriers *Enterprise* and *Hornet* in Nimitz's fleet. Nimitz also had at his disposal Boeing B-17 bombers stationed on Midway. The Japanese, unaware of the strength or position of US naval forces, sent four carriers to win air superiority in preparation for the invasion – other carriers had still not been refitted after the Coral Sea or were dispersed elsewhere.

The Japanese and US forces clashed off Midway on June 4, 1942. The result was an American victory, with all four Japanese carriers sunk for the loss of the *Yorktown*. But the Battle of Midway did not give the United States instant air or naval superiority in the Pacific. Although Japanese naval air power had suffered a severe setback, American navy fliers were still outnumbered. The loss of the *Lexington* at the Coral Sea and the *Yorktown* at Midway was followed by the sinking of the *Wasp* and *Hornet* in the fighting around Guadalcanal in the second half of 1942. During this period, the Americans had only one carrier operational in the Pacific at any given time. Some airmen found themselves without a deck to fly off and operated alongside the Marines from the precarious airstrip at Henderson Field on Guadalcanal.

The Essex-class carriers

During 1942–43, the most powerful carrier fleet in the world was taking shape in American shipyards. Back in 1940, the US Navy had been authorized to build a new generation of heavy carriers, which emerged as the Essex class. The first of these came into operation in 1943. The design of these modern ships was based on lessons learned from previous experience of carrier warfare, and great attention was paid to the practicalities of flying and servicing airplanes in a severely limited space. Much thought was also given to firefighting and damage control, with the result that Essex-class ships had a far better chance of surviving an enemy air attack than earlier carriers. Alongside

BATTLE OF MIDWAY



TORPEDO-BOMBERS

Douglas TBD Devastators (top) made up the bulk of the American torpedo-bomber force at the Battle of Midway, although half a dozen newer Grumman TBF Avengers (above) also took part. All suffered heavy losses – of the TBDs seen here on the deck of the Enterprise before the battle, only four survived.

BATTLE WAS JOINED AT MIDWAY on June 4, 1942.

The Japanese carrier aircraft opened the action with an early morning raid on the airfields on Midway Island. Japanese commanders were preparing a follow-up raid before they first became aware that there might be American carriers within striking range. By that time *Enterprise* and *Hornet* had launched their aircraft to attack the Japanese carriers. Unfortunately, the American pilots had trouble locating their target and squadrons became split up. The 15 Devastator torpedo-bombers from the *Hornet* found the carriers first, but had lost touch with their fighter cover. Slow and highly vulnerable as they flew in at low altitude to deliver their attack, the Devastators were pounced on by Zero fighters. Not a single one survived.

The aircraft from the *Yorktown* had been launched well after those from the other two carriers, but now arrived in proper formation, with fighters, dive-bombers, and torpedo-bombers prepared for a coordinated attack. The Wildcat fighters, led by Jimmy Thach, were hopelessly outnumbered by the Zeros

